Sheet 2
Finite elements method with Variational approach

1- Variational Principle Associated with Common PDEs (Prove them)

Name of equation	Partial Differential Equation (PDE)	Variational principle
Inhomogeneous wave equation	$\nabla^2\Phi + k^2\Phi = g$	$I(\Phi) = \frac{1}{2}\int_v \left[|\nabla\Phi|^2 - k^2\Phi^2 + 2g\Phi\right] dv$
Homogeneous wave equation	$\nabla^2\Phi + k^2\Phi = 0$	$I(\Phi) = \frac{1}{2}\int_v \left[|\nabla\Phi|^2 - k^2\Phi^2\right] dv$
	or	
	$\nabla^2\Phi - \frac{1}{u^2}\Phi_{tt} = 0$	$I(\Phi) = \frac{1}{2}\int^{t_o}\int_v \left[|\nabla\Phi|^2 - \frac{1}{u^2}\Phi_t^2\right] dv\, dt$
Diffusion equation	$\nabla^2\Phi - k\Phi_t = 0$	$I(\Phi) = \frac{1}{2}\int^{t_o}\int_v \left[|\nabla\Phi|^2 - k\Phi\Phi_t\right] dv\, dt$
Poisson's equation	$\nabla^2\Phi = g$	$I(\Phi) = \frac{1}{2}\int_v \left[|\nabla\Phi|^2 + 2g\Phi\right] dv$
Laplace's equation	$\nabla^2\Phi = 0$	$I(\Phi) = \frac{1}{2}\int_v \left[|\nabla\Phi|^2\right] dv$

[1] Note that $|\nabla\Phi|^2 = \nabla\Phi\cdot\nabla\Phi = \Phi_x^2 + \Phi_y^2 + \Phi_z^2$.

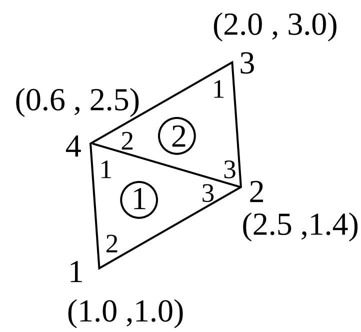
2- Resonant TM modes of a waveguide can be numerically obtained by FE method with variational approach. Prove that minimizing the functional,

$$k^2 = I(\phi) = \frac{\iint_s (\nabla\phi)^2\, dS}{\iint_s \phi^2\, ds}$$

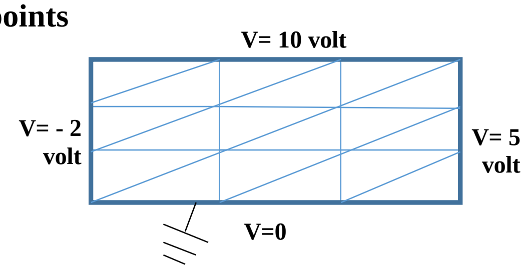
over the area *S* of the waveguide, will give the non-trivial solution to the Holmholtz equation $\nabla^2\phi + k^2\phi = 0$.

(Hint: in TM mode (E_z, H_x, H_y), $\phi = E_z$. At boundaries $\phi = 0$)

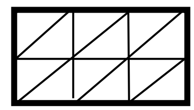
3- Find the coefficient matrix for the two element mesh shown in the figure. Given the values of the electrostatic potential at the fixed nodes 2 and 4 are $V_2 = 5$ V and $V_4 = -5$ V, find values at nodes 1 and 3.
(Region is charge free)



4- The FE method is used to get the electrostatic potential at points 12,3, and four in the hollow Region as shown in the figure. (dimensions 3x6 cm)
Get the global matrix and write the equation of solution



5- Use the shown finite element mesh to find the dominant TM-mode (TM_{11}) for the perfectly conducting waveguide axb = 3x2 cm (Hint solve Helmholtz's equation:



$$\nabla^2\phi + k^2\phi = 0 .$$

for the longitudinal component of the electric field .)